

Computation of a_2 and optimisation

The code for $f(n)$ essentially sums a geometric series. Ignoring the negligible $2^{-10^{23}}$ terms, one finds $f(n) = \sum_{k=3}^{n+2} (1/2)^k = \frac{(1/2)^3(1-(1/2)^{n+2})}{1-(1/2)} = 0.25(1-2^{-n-2})$ (by the geometric series formula ¹). Hence $a_2(n) = 2f(n) + \frac{12+0.25^{n+1}}{2 \cdot 0.25(1-2^{-n-2}) + \frac{12+4^{-n-1}}{1-0.5 \cdot 2^{-n} + 4^{-n}}}$ up to astronomically small corrections. As $n \rightarrow \infty$ this tends to 1 . In fact the tiny extra terms make a_2 just above 1 by an imperceptible amount (on the order of $2^{-10^{23}}$). In practice $a_2 \approx 1.00000\dots$ to any reasonable precision.

With $a_1 = 1$ and $a_2 \approx 1$ (slightly larger), the faster therapy is treatment 2. The constraint $a_1 x_1 + a_2 x_2 = 1$ becomes $x_1 + a_2 x_2 = 1$. To minimize $T = x_1 + x_2$ subject to this linear constraint, one checks the “corner” solutions: if $a_2 > 1$, the optimal choice is to set $x_1 = 0$ (use only the faster rate a_2); if $a_2 < 1$, one would set $x_2 = 0$. Here $a_2 > 1$, so the minimiser is $x_1 = 0, x_2 = \frac{1}{a_2}$. Numerically, since $a_2 \approx 1$, $x_2 \approx 1/1 = 1$.

Thus to two-decimal accuracy (in the ℓ^1 norm) the optimal times are $x_1 \approx 0.00, x_2 \approx 1.00$. This choice yields the minimum total time $T = x_1 + x_2 \approx 1$. (By contrast, if one had set $x_2 = 0$, then $x_1 = 1$ and $T = 1$ as well, but that would ignore the slightly higher rate of therapy 2.)

Answer: $x_1 \approx 0.00, x_2 \approx 1.00$, (to two-decimal accuracy).

Explanation: By summing the geometric series in the code we find $a_2 \approx 1$ ¹, so therapy 2 is marginally faster. Hence the minimum-time solution is to use only therapy 2: $x_1 = 0, x_2 = 1/a_2 \approx 1$. This yields total time ≈ 1 (within the required accuracy).

¹ Geometric series - Wikipedia

https://en.wikipedia.org/wiki/Geometric_series